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Trade with Nominal Rigidities: Understanding the Unemployment and Welfare Effects of the China Shock

Contents

1 Big picture

- **Question.** How do large trade shocks affect **unemployment**, **labor force participation**, and **welfare** in the short run when nominal wages cannot fall freely?
- **Setting.** The paper builds a **dynamic quantitative trade-and-migration model** with **downward nominal wage rigidity (DNWR)** and applies it to the **China shock** across US states.
- **Core challenge.** Standard quantitative trade models usually impose **full employment**. That makes them ill suited to explain the reduced-form evidence that more exposed US local labor markets saw higher **unemployment** and lower **participation** during the China shock.
- **Main methodological contribution.** The paper combines three elements:
  1. a multisector Armington trade model with input-output linkages;
  2. a dynamic migration structure with **different elasticities across sectors and across regions**;
  3. a **nominal wage floor** that can make labor demand fall short of labor supply during the transition.
- **Why it matters.** A trade shock can be *good in the long run* because it improves the terms of trade, but *painful in the short run* because nominal wages may need to fall more than rigidities allow. That changes both the employment path and the welfare accounting.
- **Main empirical question.** Can a model with DNWR quantitatively match the cross-sectional labor-market effects found by Autor, Dorn, and Hanson (2013), and what does it imply for **aggregate US welfare** and **state-level welfare**?
- **Main empirical results.**
  - The calibrated rigidity parameter is $d \approx 0.99$, so nominal wages can fall only about **1% per year** before the constraint binds.
  - The mobility parameters are $n \approx 0.54$ for movement across sectors and $k \approx 6.55$ for movement across regions.
  - In the baseline, the China shock generates aggregate US unemployment that peaks at about **1.25% in 2007** and then gradually disappears as wages adjust.
  - Aggregate labor force participation falls by up to about **1.2%** in 2007, but later reverses sign and ends up about **1% above** its preshock level.
  - US welfare still rises overall, but only by about **12 basis points** with DNWR, compared with about **31 basis points** without DNWR.
  - **30 states gain** and **20 states lose** in the baseline. Moreover, **18 states** that would have gained without DNWR instead lose once nominal rigidities are introduced.
  - In a longer-shock specification running to 2011, the aggregate welfare gain is almost extinguished and unemployment peaks around **1.75%**.
- **Economic takeaway.** The China shock is a **positive terms-of-trade shock** for the United States, but when wages are downwardly rigid it can still create substantial temporary unemploy-

ment and large regional welfare losses. Nominal rigidities therefore matter enormously for how we evaluate trade shocks.

2 Data and measurement (key objects)

2.1 Observed reduced-form evidence used to motivate the model

- The empirical benchmark is the **China-shock exposure design** of Autor, Dorn, and Hanson (2013), extended dynamically to later years.
- The paper studies how outcomes change in regions with different exposure to Chinese import competition, focusing on:
  - employment to population,
  - unemployment to population,
  - not in labor force (NILF) to population,
  - wages,
  - population.
- A crucial descriptive fact is that the China shock has **persistent employment effects** but much less persistent **unemployment effects**. By 2020, the unemployment effect is only around one-tenth of its 2007 level, while the NILF effect remains economically meaningful.
- This matters because it means persistent employment losses do *not* necessarily require persistent unemployment. They can instead come from a transition with temporary unemployment and more persistent nonparticipation.

2.2 Regions, sectors, and market-level structure in the quantitative model

- The model contains **87 regions**: **50 US states**, **36 other countries**, and a **rest-of-the-world** region.
- There are **15 sectors**:
  - **home production**,
  - **12 manufacturing sectors**,
  - **services**,
  - **agriculture**.
- Trade is sector specific and follows an **Armington structure** with iceberg trade costs and sector-specific trade elasticities.
- Production uses labor and intermediate inputs through a Cobb-Douglas technology, with input-output linkages across sectors.

2.3 Data sources for production, trade, and migration

- **US production and value added:** Bureau of Economic Analysis (BEA).
- **International input-output and bilateral trade data:** World Input-Output Database (WIOD).
- **Within-US manufacturing trade flows:** WIOD combined with the Commodity Flow Survey.
- **US state–foreign-country trade flows:** merchandise trade statistics.
- **Services trade flows:** Regional Economic Accounts of the BEA, WIOD, and bilateral distances under a gravity structure.
- **Agriculture trade flows:** agricultural census, National Marine Fisheries Service census, and WIOD.
- **Migration across US sector-states:** CPS, ACS, IRS state-to-state migration data, and BLS sector-state employment data.

The resulting migration matrix is built so that total cross-state movements match the IRS data and stock changes across sector-state pairs match the census and BLS data.

2.4 Key calibration objects

- The China shock is modeled as a sequence of **productivity shocks in Chinese manufacturing sectors only**.
- The shock is decomposed multiplicatively as

$$\widehat{A}_{\text{China},s,t} = \widehat{A}^{1}_{\text{China},t}\widehat{A}^{2}_{\text{China},s},$$

  so there is a common time component and a common sector component.
- The **baseline** shock lasts from **2001 to 2007**. An alternative specification lets it last until **2011**.
- The key parameters d, n, and k are chosen to match three reduced-form moments from Autor, Dorn, and Hanson (2013):
  - the unemployment response,
  - the NILF response,
  - the population response.
- The paper also sets the trade elasticity to **6**, the discount factor to **0.95**, and the unemployment-insurance/risk-sharing parameter to **0.5**.

2.5 Timing and economic environment

- Workers are **forward looking** and each period choose in which sector-region to participate next period.
- Their decision depends on:

- current flow utility,
- moving costs,
- expected continuation values,
- unemployment risk when DNWR binds.

- The model starts from a year-2000 benchmark with **full employment**. This is interpreted as a reasonable approximation because 2000 was the peak of a business cycle with very low unemployment.

3 Models and empirical specifications (all equations + interpretation)

3.1 Reduced-form empirical specifications motivating the model

The dynamic reduced-form specification extends Autor, Dorn, and Hanson (2013):

$$\Delta Y_{i,t+h} = \alpha_t + \beta_{1h}\,\Delta IP^{cu}_{i,t} + X'_{i,t}\beta_2 + \varepsilon_{i,t+h}.$$

- $\Delta Y_{i,t+h}$: 10-year-equivalent change in an outcome for commuting zone i.
- $\Delta IP^{cu}_{i,t}$: growth in exposure to Chinese import competition.
- $X_{i,t}$: standard controls from Autor, Dorn, and Hanson (2013).

Interpretation. This regression traces out how the effect of the China shock evolves over time. The paper shows that employment effects remain persistent, but unemployment effects fade much faster than NILF effects.

To connect those facts to nominal rigidity, the paper also estimates:

$$\Delta U_{i,t+h} = \gamma_t + \beta_{1,h}\,\Delta IP^{cu}_{i,t} + \beta_{2,h}\,\mathrm{Rig}_{s(i),t} + \beta_{3,h}\,\mathrm{Rig}_{s(i),t} \times \Delta IP^{cu}_{i,t} + X'_{i,t}\beta_4 + \varepsilon_{i,t+h},$$

where $\mathrm{Rig}_{s(i),t}$ is a preshock state-level proxy for DNWR.

Interpretation. If DNWR matters, regions with more rigid wages should experience a larger unemployment response to the same trade shock. That is exactly what the paper finds.

3.2 Production, prices, and trade

The price of sector s goods from region i sold in region j is

$$p_{ij,s,t} = \tau_{ij,s,t} A^{-1}_{i,s,t} W^{f_{i,s}}_{i,s,t} \prod_{r=1}^{S} P^{f_{i,rs}}_{i,r,t}.$$

The sectoral price index is

$$P^{1-\sigma_s}_{j,s,t} = \sum_{i=1}^{I} p^{1-\sigma_s}_{ij,s,t}.$$

Market clearing for sector s in region i is

$$R_{i,s,t} = \sum_{j=1}^{I} \lambda_{ij,s,t} \left[a_{j,s} \left(\sum_{r=1}^{S} W_{j,r,t} L_{j,r,t} + D_{j,t} \right) + \sum_{r=1}^{S} f_{j,sr} R_{j,r,t} \right],$$

with trade shares

$$\lambda_{ij,s,t} = \frac{p_{ij,s,t}^{1-\sigma_s}}{\sum_{m=1}^{I} p_{mj,s,t}^{1-\sigma_s}}.$$

Labor demand satisfies

$$W_{i,s,t} L_{i,s,t} = f_{i,s} R_{i,s,t}.$$

Interpretation. This block is a standard multisector trade model with input-output linkages. China's productivity growth changes prices, trade shares, revenues, and labor demand across all regions and sectors.

3.3 Worker mobility and nested labor supply

Workers solve a dynamic location-sector choice problem:

$$v_{j,s,t} = U_{j,s,t} + \max_{i,r} \left\{ \beta E\left[v_{i,r,t+1}\right] - J_{ji,sr} + \varepsilon_{i,r,t} \right\}.$$

The idiosyncratic shocks are nested Gumbel, which yields different movement elasticities across sectors and regions. Expected lifetime utility is

$$V_{j,s,t} = U_{j,s,t} + \ln \left(\sum_{i=1}^{I} \left[\sum_{r=0}^{S} \exp \left(\frac{\beta V_{i,r,t+1} - J_{ji,sr}}{n} \right) \right]^{n/k} \right)^{k} + \gamma k,$$

where $1/n$ is the elasticity of mobility across sectors and $1/k$ is the elasticity across regions.

Conditional sector choice within a region is

$$m_{ji,sr,t}^{i} = \frac{\exp\left((\beta V_{i,r,t+1} - J_{ji,sr})/n \right)}{\sum_{h=0}^{S} \exp\left((\beta V_{i,h,t+1} - J_{ji,sh})/n \right)},$$

and the probability of moving from (j, s) to region i is

$$m_{ji,s\#,t} = \frac{\left[\sum_{h=0}^{S} \exp\left((\beta V_{i,h,t+1} - J_{ji,sh})/n \right) \right]^{n/k}}{\sum_{m=1}^{I} \left[\sum_{h=0}^{S} \exp\left((\beta V_{m,h,t+1} - J_{jm,sh})/n \right) \right]^{n/k}}.$$

Hence the total movement share is

$$m_{ji,sr,t} = m_{ji,sr,t}^{i} \, m_{ji,s\#,t},$$

and next period participation evolves as

$$\ell_{i,r,t+1} = \sum_{j=1}^{I} \sum_{s=0}^{S} m_{ji,sr,t} \, \ell_{j,s,t}.$$

Interpretation. This is the paper's second major innovation relative to Caliendo, Dvorkin, and Parro (2019): workers can be relatively flexible across sectors but much less flexible across regions. That helps match the data on NILF and population responses.

3.4 Flow utility, unemployment risk, and home production

Without DNWR, labor markets clear and workers in market sectors receive flow utility equal to the log real wage:

$$U_{i,s,t} = \ln(q_{i,s,t}), \qquad q_{i,s,t} = \frac{W_{i,s,t}}{P_{i,t}},$$

with aggregate consumption price index

$$P_{i,t} = \prod_{s=1}^{S} P_{i,s,t}^{a_{i,s}}.$$

With DNWR, labor demand can be below labor supply, so the employment probability is

$$p_{i,s,t} = \frac{L_{i,s,t}}{\ell_{i,s,t}}.$$

Expected real income is

$$q_{i,s,t} = p_{i,s,t}\frac{W_{i,s,t}}{P_{i,t}},$$

and flow utility becomes

$$U_{i,s,t} = \ln\left(\Delta_{i,s,t} q_{i,s,t}\right),$$

where

$$\Delta_{i,s,t} = z^{1-p_{i,s,t}}\left(\frac{1 - z(1 - p_{i,s,t})}{p_{i,s,t}}\right)^{p_{i,s,t}} \leq 1.$$

Home production utility is

$$U_{i,0,t} = \ln(q_{i,0,t}),$$

where $q_{i,0,t}$ is exogenous and time invariant.

Interpretation. The term $\Delta_{i,s,t}$ is a risk penalty from possible unemployment. This creates a mechanism through which higher unemployment reduces participation, helping the model match the NILF response in the data.

3.5 Downward nominal wage rigidity and the nominal anchor

The labor market may fail to clear:

$$L_{i,s,t} \leq \ell_{i,s,t}.$$

Nominal wages cannot fall below a fraction of last period's wage:

$$W_{i,s,t} \geq d_{i,s} W_{i,s,t-1}, \qquad d_{i,s} \geq 0.$$

The DNWR condition holds with complementary slackness:

$$(\ell_{i,s,t} - L_{i,s,t})\big(W_{i,s,t} - d_{i,s}W_{i,s,t-1}\big) = 0.$$

The model is closed with a nominal anchor:

$$\sum_{i=1}^{I}\sum_{s=1}^{S} W_{i,s,t}L_{i,s,t} = g\sum_{i=1}^{I}\sum_{s=1}^{S} W_{i,s,t-1}L_{i,s,t-1}.$$

Interpretation. The wage floor prevents the nominal wage from adjusting down fast enough after the China shock. The nominal anchor prevents the rest of the world from inflating away the rigidity. Together, they generate temporary unemployment.

3.6 Equilibrium, dynamic hat algebra, and welfare

A temporary equilibrium at time t is a set of wages and employment levels satisfying the price, trade, labor-demand, DNWR, and nominal-anchor conditions given last period's nominal variables and current labor supply.

A sequential equilibrium combines those temporary equilibria with workers' dynamic movement decisions over time.

For counterfactuals, the paper uses **dynamic exact hat algebra**. Rather than estimate full technology levels and iceberg trade costs period by period, it expresses the equilibrium in growth rates and deviations from the year-2000 benchmark.

The China-shock exposure measure used in calibration is

$$\text{Exposure}_i = \sum_{s=1}^{S} \frac{L_{i,s,2000}}{L_{i,2000}} \frac{\widehat{\Delta X}^{2007-2000}_{C,US,s}}{R_{US,s,2000}}.$$

Interpretation. This is a Bartik-style measure: regions more specialized in sectors where Chinese import growth is stronger are more exposed.

Welfare is computed as a permanent-equivalent variation in expected lifetime utility for workers initially located in sector-region (j, s). It aggregates changes in:

- risk-adjusted real income,
- within-region sector mobility options,
- across-region mobility options.

Interpretation. Welfare is not just about wages today. It includes future wages, unemployment risk, and the option value of moving across sectors and states.

3.7 Calibration mapping

The China shock is calibrated so that model-implied US imports from China match predicted changes from reduced-form regressions that use other high-income countries' imports from China as instruments.

The parameters d, n, and k are then chosen so that the model matches three key reduced-form responses from Autor, Dorn, and Hanson (2013):

- a \$1,000 per worker increase in exposure raises unemployment/population by **0.22 percentage points**;
- raises NILF/population by **0.55 percentage points**;
- lowers population by **5 basis points**.

Interpretation. The model is not calibrated to match everything. Instead, it is disciplined by a small number of moments that are central to the China-shock literature.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard trade approach

- Standard quantitative trade models typically impose **full employment**, so they cannot directly speak to unemployment spikes after trade shocks.
- A single mobility elasticity across sectors and regions tends to produce too little spatial dispersion in employment and income effects.
- Without nominal rigidity, the usual exposure measure in Autor, Dorn, and Hanson (2013) is not enough to summarize short-run employment effects, because wages would adjust and the economy would primarily respond through real reallocation and consumer prices.

4.2 What identifies the new approach

- **The time path and sector pattern of the China shock** are disciplined by changes in imports from China predicted from other high-income countries.
- **The DNWR parameter** d is identified from how much unemployment has to rise when wages cannot fall quickly enough.
- **The sector mobility parameter** n is identified from how strongly participation/NILF responds.
- **The regional mobility parameter** k is identified from how much population reallocation occurs across places.

Key insight. The paper maps three reduced-form regional coefficients into three structural parameters. This is what lets the model speak quantitatively to the Autor-Dorn-Hanson evidence.

4.3 Why DNWR is empirically plausible here

- Micro evidence from recent labor studies shows strong downward nominal wage rigidity for both continuing workers and new hires.
- The China-shock data show that persistent employment losses are driven much more by **NILF** than by **unemployment**, so a model with transitory unemployment is not contradicted by the long-run evidence.

- Regions with stronger preshock DNWR experienced larger unemployment increases from the China shock. In 2007, highly rigid commuting zones experienced about an additional **0.17 percentage point** rise in unemployment/population.

4.4 Empirical credibility of the quantitative application

- The model matches not only the three targeted moments, but also **untargeted** changes in manufacturing and nonmanufacturing employment and wages fairly well.
- The model reproduces spatial dispersion in employment and income changes better than earlier quantitative trade models.
- The state-level standard deviation of model-implied employment changes is **1.11**, close to the **1.18** implied by the empirical reduced-form estimates. For income per capita, the model gives **2.1** versus **1.9** in the data.

4.5 Nominal assumptions and alternative interpretations

- The paper uses a simple nominal anchor—constant growth of world nominal GDP in US dollars—to make the model solvable.
- The authors argue that similar qualitative results would arise if China prevented renminbi appreciation through exchange-rate and monetary policy while the United States did not fully offset the resulting demand deficiency.
- The paper is transparent that this nominal side is stylized; the point is to isolate the role of DNWR in a rich regional trade environment.

4.6 Relation to the literature

- Relative to Caliendo, Dvorkin, and Parro (2019), the paper adds DNWR and separates sectoral from spatial mobility.
- Relative to search-and-matching trade models, the paper emphasizes **nominal wage adjustment** rather than vacancy posting or matching frictions as the main source of short-run unemployment.
- Relative to Eaton, Kortum, and Neiman (2013), the paper studies a rich multisector, multiregion trade shock instead of sudden stops across countries.

5 Tables and figures

5.1 Figure 1: Dynamic reduced-form effects on employment, unemployment, and NILF

Read:

- Employment effects of the China shock remain negative and significant for a long time.
- Unemployment effects fade much faster and become economically small by 2020.

- NILF effects remain persistent.

Punchline. The persistence in employment losses does not imply permanently high unemployment. It is consistent with a model in which unemployment is temporary but some workers exit the labor force for longer.

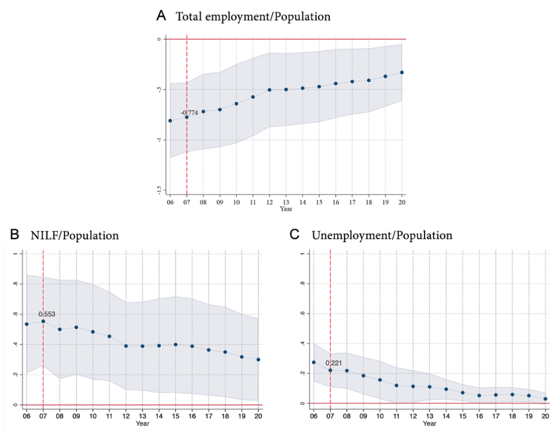

FIG. 1.—Effects of China shock on employment and nonemployment. The figure reports two-stage least squares coefficient estimates for β_{1h} in equation (1) and 95% confidence intervals for these estimates. Each coefficient comes from a separate instrumental variable regression following equation (1). Each regression stacks the change in the specified outcome between 1990 and 2000 and between 2000 and the year indicated on the horizontal axis. The coefficients for 2007 (dashed vertical line) match those from table 5 in Autor, Dorn, and Hanson (2013).

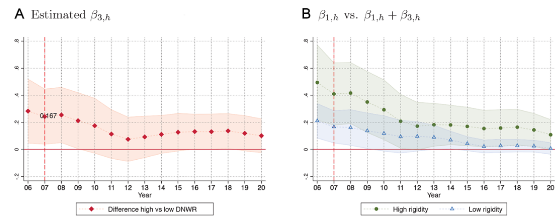

FIG. 2.—China shock and unemployment in commuting zones with high versus low DNWR. The figure reports two-stage least squares coefficient estimates in equation (2) and 95% confidence intervals for these estimates. Coefficients in each year come from a separate instrumental variable regression following equation (2), where the measure $\mathrm{Rig}_{s(i),r}$ is a dummy taking a value of 1 in commuting zone i if the share of individuals with negative wage changes in state s is below the mean across all states.

5.2 Figure 2: DNWR and cross-sectional unemployment responses

Read:

- Regions with more stringent preshock DNWR see a larger unemployment response to the China shock.
- In 2007, high-rigidity commuting zones experience roughly an additional **0.17 percentage point** increase in unemployment/population relative to low-rigidity zones.

Punchline. DNWR is not just a theoretical add-on. It lines up with the cross-sectional evidence.

5.3 Table 1: Calibration, model fit, welfare, and alternative specifications

Read:

- Baseline calibrated parameters: $n = 0.537$, $k = 6.548$, $d = 0.991$.
- A \$1,000 worker exposure increase implies: unemployment +0.221, NILF +0.553, manufacturing employment −0.605, nonmanufacturing employment −0.169 in decadalized population-share terms.
- Population falls by 0.050%, manufacturing wages are roughly flat, and nonmanufacturing wages fall by about 1.177%.
- Mean welfare gain is **0.126%** (about 12.6 basis points) in the baseline, versus **0.312%** without DNWR.

- In the longer-shock case, mean welfare gain drops to **0.011%**; with no migration it is **0.138%**; imposing $n = k$ yields too much population reallocation.

Punchline. The model is tightly disciplined by data and the baseline welfare gains are dramatically smaller once nominal rigidities are present.

TABLE 1
EMPLOYMENT, POPULATION, WAGE, AND WELFARE EFFECTS OF EXPOSURE TO CHINA ACROSS US REGIONS AND ASSOCIATED PARAMETERS GENERATING THEM

	Autor, Dorn, and Hanson (2013) (1)	Baseline (2)	Longer (3)	No Migration (4)	$\nu = \kappa$ (5)
	A. Change in Population Shares				
1. Unemployment (targeted)	.221*	.221	.221	.221	.221
2. NILF (targeted)	.553*	.553	.553	.553	.553
3. Manufacturing employment	−.596*	−.605	−.578	−.602	−.613
4. Nonmanufacturing employment	−.178	−.169	−.196	−.172	−.161
	B. Percentage Changes				
5. Population (targeted)	−.050	−.050	−.050	.000	−.211
6. Manufacturing wage	.150	.023	.209	.016	.039
7. Nonmanufacturing wage	−.761*	−1.177	−.966	−1.204	−1.182
	C. Welfare				
8. Welfare vs. exposure		−.091	−.134	−.081	−.099
9. Mean welfare change		.126	.011	.138	.124
10. Mean welfare change no DNWR		.312	.450	.314	.313
	D. Parameters				
ν		.537	.706	.611	.606
κ		6.548	13.53		.606
δ		.991	.994	.990	.991

NOTE.—The changes for the first four coefficients are measured from 2000 to an average of 2006–8, multiplied by 10/7 to turn into decadal changes. Population and wages are simply measured in percentage change (between 2000 and 2006–8), still turned into decadal changes. Welfare is obtained as described at the end of sec. III.F. ν governs substitution between sectors, κ governs substitution between regions, and δ governs the DNWR. Column 1 reproduces the Autor et al. (2013) results from their tables 4 (panel C, col. 1), 5 (panel B, row 1), and 7 (panel B, cols. 1 and 4). Column 2 gives the results in our baseline specification. Column 3 describes a longer shock that lasts until 2011 instead of until 2007. Column 4 eliminates migration across US states. Column 5 imposes $\nu = \kappa$. In col. 4, κ is not reported because this parameter is irrelevant without migration.
* $p < .01$.

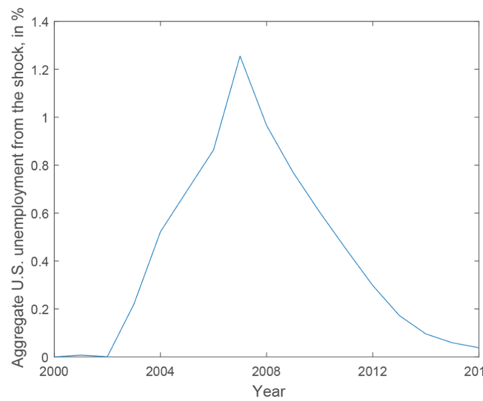

FIG. 3.—Path of aggregate US unemployment generated by China shock in baseline specification between 2000 and 2016.

5.4 Figure 3: Aggregate unemployment path in the baseline

Read:

- Aggregate unemployment generated by the China shock rises gradually and peaks at about **1.25% in 2007**.
- It then declines to near zero by **2016** as wages slowly adjust downward.

Punchline. DNWR creates a medium-run unemployment hump rather than a permanent rise in unemployment.

5.5 Figure 4 and Figure 5: Welfare across states and sector-states

Read:

- Figure 4 shows that states with greater China exposure tend to have lower welfare gains.
- There are **30 gaining states** and **20 losing states** in the baseline.
- Figure 5 shows much wider variation at the sector-state level than at the state level.
- Welfare ranges from about **-60 to +156 basis points** across sector-states, compared with about **-31 to +99 basis points** across states.

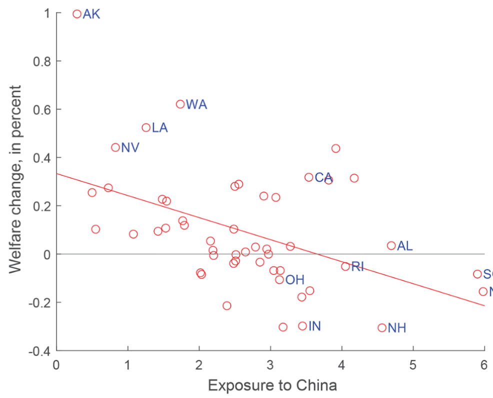

FIG. 4.—Welfare change versus exposure to China across US states in baseline specification. Selected states are labeled with two-letter abbreviations.

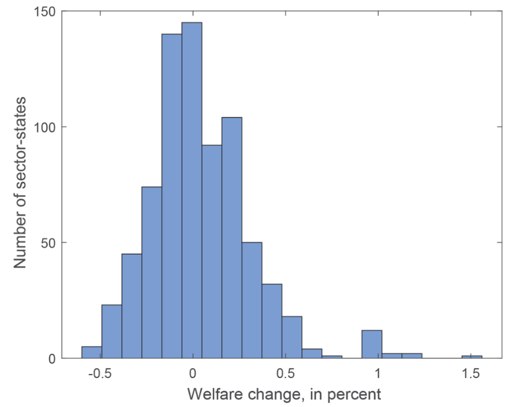

FIG. 5.—Histogram of welfare changes across different US sector-states in baseline specification.

Punchline. Aggregate US gains conceal large distributional heterogeneity across places and sectors.

5.6 Figure 6 and Figure 7: Longer-shock specification

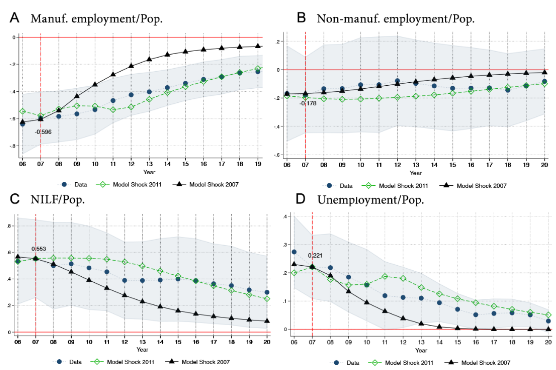

FIG. 6.—Effects of China shock on employment and nonemployment. Lines with circles show the two-stage least squares coefficient estimates for β_{1h} in equation (1), and shaded areas represent their 95% confidence intervals. These coefficients are the same as in figure 1 with the exception that the total employment to population effects in figure 1*A* are separated into manufacturing to population (*A*) and nonmanufacturing to population (*B*). Lines with diamonds display the effects in our model when the China shock lasts between 2001 and 2011 ("Model Shock 2011"), while lines with triangles display the effects in our baseline model when the China shock lasts between 2001 and 2007 ("Model Shock 2007"). The three lines coincide in 2007 for *C* and *D* by construction.

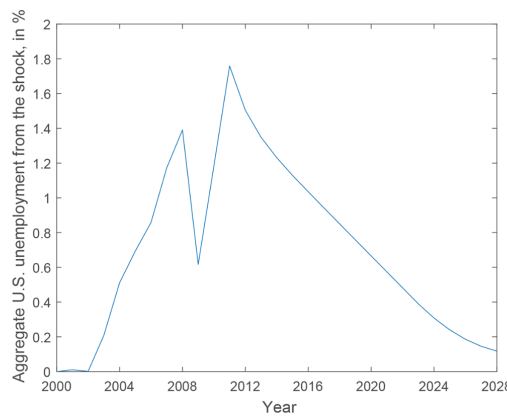

FIG. 7.—Path of aggregate US unemployment generated by China shock in longer-shock specification between 2000 and 2028.

Read:

- Extending the China shock to 2011 improves the model's fit to the dynamic persistence of the empirical cross-sectional responses.
- The longer-shock unemployment path peaks at about **1.75%** and lasts much longer.
- The longer-shock specification brings aggregate US welfare close to zero.

Punchline. The more persistent the shock, the more the short-run nominal rigidities destroy the long-run gains.

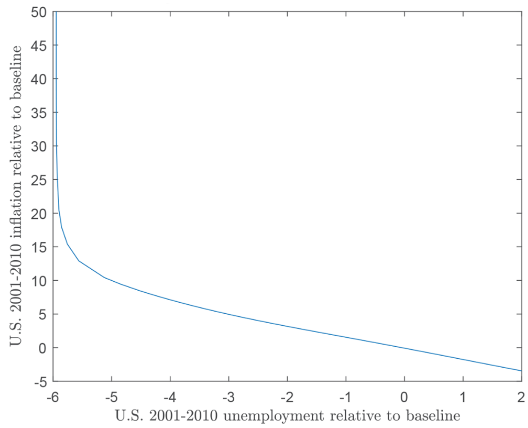

FIG. 8.—Aggregate US unemployment between 2001 and 2010 (in year-points) relative to baseline value (*x*-axis) and aggregate US inflation between 2001 and 2010 (in year-points) relative to baseline value (*y*-axis). The figure provides a notion of the sacrifice ratio between unemployment and inflation that is implicit in the model.

5.7 Figure 8: The model-implied sacrifice ratio

Read:

- The cumulative unemployment effect from 2001 to 2010 is about **6 year-points** in the baseline.
- Near the baseline, reducing unemployment by **1 year-point** requires about **1.63 year-points** more inflation.
- If unemployment is pushed 3 year-points below baseline, the cost rises to about **2.2 year-points** of inflation.
- As unemployment approaches zero, the sacrifice ratio becomes extremely large.

Punchline. Monetary accommodation can offset the employment effects of the China shock, but the inflation trade-off becomes steep.

6 Short summary

- **Method.** Dynamic quantitative trade-and-migration model with downward nominal wage rigidity and separate sectoral and regional mobility elasticities.
- **Identification.** Calibrate the China shock using imports-from-China data and pin down d, n, and k with the Autor-Dorn-Hanson unemployment, NILF, and population moments.
- **Application.** China shock across US states from 2000 onward.
- **Main result.** The China shock still raises aggregate US welfare, but DNWR cuts those gains by roughly two-thirds and creates substantial short-run unemployment and large regional welfare losses.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- A trade shock that is beneficial in the long run can generate substantial short-run unemployment when nominal wages are downwardly rigid.
- A model with DNWR can match the main reduced-form labor-market facts of the China shock much better than standard full-employment quantitative trade models.
- Regional welfare assessments change materially once short-run labor-market dislocation is taken seriously.

7.2 Economic intuition

- China's productivity growth improves US terms of trade, which tends to increase the US real wage in the long run.
- But the same shock also lowers the price of US exports, which requires a fall in nominal wages to keep labor markets clearing.
- If nominal wages cannot fall fast enough, employment falls instead.
- Unemployment risk then discourages labor-force participation, magnifying the employment effect.
- Over time, as the nominal wage slowly adjusts, unemployment disappears and the favorable terms-of-trade effect dominates.

7.3 Takeaway

This paper's main lesson is that **trade shocks should not be evaluated with long-run real-allocation logic alone**. Once downward nominal wage rigidity is introduced, the same China shock that raises US welfare on average can also generate large transitional unemployment, persistent nonparticipation, and meaningful welfare losses in the most exposed places. The paper therefore gives a clear quantitative reason to put **nominal adjustment frictions** at the center of trade-shock analysis.